

IWOAT Summer School 2026

An Introduction to Stable Homotopy Theory and Brave New Algebra

Date: August 9–16, 2026

Venue: Lecture hall, Siyuan building

Lunch: Wuke hotel, 3rd floor

Schedule at a Glance

Time	Sunday, Aug. 9	Monday, Aug. 10	Tuesday, Aug. 11	Wednesda Aug. 12	Thursday, Aug. 13	Friday, Aug. 14	Saturday, Aug. 15	Sunday, Aug. 16
8:45–9:15	Arrival, Registra- tion	Registration					Free dis- cussion	Departure
9:15–9:30		Opening remarks						
9:30–10:30		Andrew J. Blumberg	Shangjie Zhang	Runji Li	Courtney Hauf	Aowen Fan		
10:30–11:00		Tea break	Tea break	Tea break	Tea break	Tea break		
11:00–12:00		Yueshi Hou	Shangjie Zhang	Alex Waugh	Xiansheng Li	Xiansheng Li		
12:00–2:00		Lunch	Lunch	Lunch	Lunch	Lunch		
2:00–3:00		Yueshi Hou	Yuxuan Li	Alex Waugh	Shuhan Zheng	Sihao Ma		
3:00–3:30		Tea break	Tea break	Tea break	Tea break	Tea break		
3:30–4:30		Shuhan Zheng	Yifan Jin	Stephanie Oh	Yuchen Wu	Yuchen Wu		

Monday, August 10

Time	Talk / event	Speaker
8:45–9:15	Registration	
9:15–9:30	Opening remarks	
9:30–10:30	Overview (online)	Andrew J. Blumberg
10:30–11:00	<i>Tea break</i>	
11:00–12:00	Model categories	Yueshi Hou
12:00–2:00	<i>Lunch</i>	
2:00–3:00	Simplicial (model) categories	Yueshi Hou
3:00–3:30	<i>Tea break</i>	
3:30–4:30	$(\infty,1)$-categories	Shuhan Zheng

Tuesday, August 11

Time	Talk / event	Speaker
9:30–10:30	Diagram spectra; symmetric, orthogonal, excisive functors	Shangjie Zhang
10:30–11:00	<i>Tea break</i>	
11:00–12:00	Γ-spaces	Shangjie Zhang
12:00–2:00	<i>Lunch</i>	
2:00–3:00	Stable model categories, SM7 revisited	Yuxuan Li
3:00–3:30	<i>Tea break</i>	
3:30–4:30	Spectra and excisive functors in $(\infty,1)$-categories	Yifan Jin

Wednesday, August 12

Time	Talk / event	Speaker
9:30–10:30	The model structures on rings and modules in diagram spectra	Runji Li
10:30–11:00	<i>Tea break</i>	
11:00–12:00	Morita theory, module categories, bar constructions and cyclic bar constructions	Alex Waugh
12:00–2:00	<i>Lunch</i>	
2:00–3:00	Operads	Alex Waugh
3:00–3:30	<i>Tea break</i>	
3:30–4:30	Eilenberg–Mac Lane spectra, HM-modules, and chain complexes	Stephanie Oh

Thursday, August 13

Time	Talk / event	Speaker
9:30–10:30	Units and Thom spectra	Courtney Hauf
10:30–11:00	<i>Tea break</i>	
11:00–12:00	THH and TAQ; construction, definition, basic properties	Xiansheng Li
12:00–2:00	<i>Lunch</i>	
2:00–3:00	Algebraic K-theory; inputs as Waldhausen categories, stable categories	Shuhan Zheng
3:00–3:30	<i>Tea break</i>	
3:30–4:30	Picard and Brauer groups, generalized Thom spectra	Yuchen Wu

Friday, August 14

Time	Talk / event	Speaker
9:30–10:30	Localization and local objects	Aowen Fan
10:30–11:00	<i>Tea break</i>	
11:00–12:00	Localization in the stable setting, stable ∞-categories and localization, symmetric monoidal structures	Xiansheng Li
12:00–2:00	<i>Lunch</i>	
2:00–3:00	A brief introduction to chromatic localizations and the chromatic perspective on the stable category (online)	Sihao Ma
3:00–3:30	<i>Tea break</i>	
3:30–4:30	Visions of spectral algebraic geometry	Yuchen Wu